

Numerical Analysis homework 1

Zirui Chen 22435036 *

(24), Zhejiang University

Due time: December 23, 2024

I

$$477 = (101001101)_2$$

II

$$\frac{3}{5} = (1.0011001 \dots)_2 \times 2^{-1}$$

III

Prove:

Because $x = 1.0 \times \beta^e$, $x_R = 1.0 \dots 01 \times \beta^e$, $x_L = 0.1 \dots 11 \times \beta^{e-1}$, so, $x_R - x = \beta^{e-(p-1)}$, $x_L - x = \beta^{e-1-(p-1)}$, $x_R - x = \beta(x - x_L)$.

□

IV

IV-a

IEEE754(2, 24, -126, 127)

$$x_R = 1.001101 \dots 1010 \times 2^{-1}$$

$$x_L = 1.001101 \dots 1001 \times 2^{-2}$$

$$x_R - x = \frac{2}{5} \times 2^{-24}, x_L - x = \frac{3}{5} \times 2^{-24}, \text{ so we should choose } x_R. \text{Rel}(\frac{3}{5}) = \frac{\|x_R - x\|}{\|x\|} = \frac{2}{3} \times 2^{-24}$$

V

Using nearest rounding, error will not be more than half of the machine precision plus β^e . if we drop excess bits, the unit roundoff will be the same as the machine precision plus β^e .

VI

According to the Theorem 4.49, set $x = 1, y = \cos \frac{1}{4}, \beta = 2$ then $1 - \frac{y}{x} = (0.03108 \dots)_2 = (0.000001)_{10}$ So, at least 6 bits of precision are lost.

VII

We can use Taylor expansion and double angle formula.

*Electronic address: 18107732576@163.com

VIII

$$Cond_f(x) = \frac{\|x\| \cdot \|\nabla f\|}{\|f(x)\|} \quad (\text{a}) \quad (x-1)^a \Rightarrow \left\| \frac{ax}{x-1} \right\|, x \rightarrow 1 \quad (\text{b}) \quad \ln x \Rightarrow \left\| \frac{1}{\ln x} \right\|, x \rightarrow 1 \quad (\text{c}) \quad e^x \Rightarrow \|x\|, x \rightarrow \infty \quad (\text{d}) \quad \arccos x \Rightarrow \left\| \frac{x}{\sqrt{1-x^2} \cdot \arccos x} \right\|, x \rightarrow \pm 1$$

IX

IX-a

$$cond_f(x) = \left| \frac{x \cdot e^{-x}}{1-e^{-x}} \right| = \left| \frac{x}{e^x-1} \right| \text{ it is trival to show that } cond_f(x) \leq 1 \text{ for } x \in (0, 1).$$

IX-b

According to Theroem4.40, we have,

$$f_A(x) = fl(1 - fl(e_{-x})) \quad (1)$$

$$= (1 - e^x(1 + \delta_1))(1 + \delta_2) \quad (2)$$

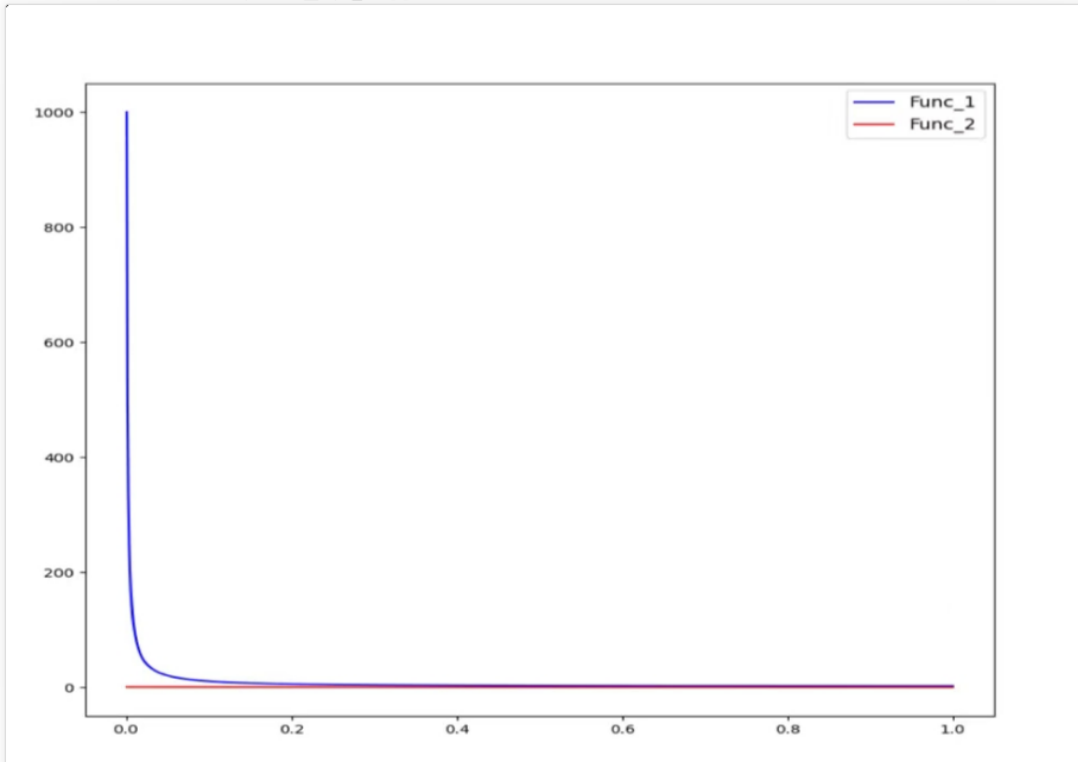
$$= (1 - e^{-x})(1 + \delta_2 - \delta_1 \frac{1}{e^x - 1}) \quad (3)$$

$$(4)$$

$$\text{where } \delta_1, \delta_2 \leq \varepsilon_u, \text{ set } \varphi(x) = \frac{1-2e^{-x}}{1-e^{-x}}$$

$$\text{so, } cond_A(x) \leq \frac{\varphi(x)}{cond_f(x)} = \frac{e^x+2e^{-x}-3}{x(1-e^{-x})}$$

IX-c



X

XI

Prove:

$$cond_f(x) = \left\| \left[\frac{a_0}{r} \frac{\partial r}{\partial a_0}, \dots, \frac{a_{n-1}}{r} \frac{\partial r}{\partial a_{n-1}} \right] \right\|_1$$

$$\frac{\partial r}{\partial a_k} = r^k + \frac{\partial r}{\partial a_k} q'(r) \Rightarrow \frac{\partial r}{\partial a_k} = -\frac{r^k}{q'(r)}$$

Because the 1-norm and the infinity norm are equivalent. consider the 1-norm, when $q(x) = \prod_{k=1}^p (x - k)$, we have,
 $cond_q(r) = \frac{r^{n-1}}{\prod_{k=1, k \neq r}^n |r - k|}$ So when $r = n$. $cond_p(r) \rightarrow \infty$

□

XII

Set $(2, 2, -1, 1)$, $a = 1.0 \times 2^0$, $b = 1.1 \times 2^0$, $\frac{a}{b} = \frac{10}{11}$

$$M = fl_{register}(\frac{1.0}{1.1}) = fl_{register}((0.1010 \dots)_2) = (0.101)_2$$

and $fl(1.01 \times 2^{-1}) = 1.0 \times 2^{-1}$, so $e_{rel} = \frac{9}{20} > \frac{1}{2} \varepsilon_M = \frac{1}{4}$

XIII

$128 = 1.0 \dots 0 \times 2^7$, $129 = 1.0 \dots 01 \times 2^7$ when $x_1, x_2 \in [128, 129]$, $x_2 - x_1 = \varepsilon_M \cdot 2^7 = 2^{-16} > 10^{-6}$ so we cant compute the root with absolute accuracy $< 10^{-6}$.

XIV

when $x_{i+1} - x_i$ is much smaller than those of other adjacent pairs. the coefficient matrix of the interpolating polynomial is ill-conditioned.

Acknowledgement

Give your acknowledgements here(if any).